Sandeep Malik

AI Architect | Principal Engineer — Enterprise Agent Platforms, Harness Engineering & MCP
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SUMMARY

AI Architect and Principal Software Engineer with 14+ years designing enterprise platforms across AI, data, and distributed systems. Hands-on builder of production agent platforms, MCP integrations, secure RAG applications, and AI-assisted developer workflows. At Autodesk, leads a central agent platform and has shipped 3 production agents, 5 MCPs, and 2 IDE plugins; built an API modernization agent for 1,500+ applications with greater than 85% success by code complexity. Combines recent AI architecture with deep cloud, data, and enterprise application experience (handled up to ~2 PB and ~200K events/sec).

CORE SKILLS

Agentic AI & Architecture: Enterprise AI solutions, agent harnesses and runtimes, multi-agent orchestration, MCP interoperability, tool integration, context assembly, RAG, deterministic workflows, human oversight
AI Quality & Trust: Agent evaluation, LLM-as-judge, guardrails, AI observability, secure coding, RBAC, vulnerability and compliance data, production monitoring
Models & Frameworks: LangGraph, Google ADK, LangChain, LlamaIndex, Azure OpenAI, AWS Bedrock, Pinecone, Elasticsearch, OpenSearch
Cloud, Data & Engineering: Python, Java, Scala, SQL, Node.js, Snowflake, Databricks, Spark, Kafka, PostgreSQL, AWS, Azure, Kubernetes, Docker, Terraform, CI/CD

EXPERIENCE

Principal Software Engineer / Technical Lead, Autodesk, Trust AI Engineering

Jul 2025 – Present | Montreal, Canada

- Lead architecture and delivery for a central BYO-agent / BYO-MCP platform — agent and MCP catalog, discovery, orchestration, worker harness, and engineering standards; shipped 3 production agents (1 proactive, 2 reactive), 5 MCPs, and 2 IDE plugins used across Autodesk.
- Designed and built a reusable worker system and agent runtime so teams compose agent and MCP capabilities through shared patterns — environment-aware endpoints, common authentication, typed outputs, guards, and observability.
- Built a proactive API modernization agent that migrates legacy, vulnerability-prone APIs to GraphQL across 1,500+ internal applications — direct and dependency-based usage in Python, Go, Java, Ruby on Rails, and Perl — at greater than 85% success by code complexity, with a UI progress dashboard.
- Built an MCP server giving developers in-context checks against cloud and secure-coding standards (RAG, vector search, observability), shifting security feedback earlier in the SDLC; delivered a multi-step threat-modeling MCP workflow as a reusable security capability.
- Designed ingestion pipelines and a PostgreSQL schema for vulnerability and compliance data, enabling reliable, access-controlled MCP and agent workflows.

Data Architect / Staff Engineer, Databook Labs

Dec 2022 – Jul 2025 | Montreal, Canada (Remote)

- Built the company data platform from the ground up and delivered its first semantic layer (early 2023), powering Databook AI.
- Led a GPT product for strategic sales using multi-query and step-back retrieval, reranking, and LLM-as-judge to improve grounding and personalization.
- Designed agentic workflows with LangGraph and Google ADK and a semantic index across Pinecone, Elasticsearch, Snowflake, and Databricks; led a team of data engineers.

Principal Data Engineer, Sitecore

Mar 2022 – Dec 2022 | Montreal, Canada

- Led the core Search team supporting AI-backed search and e-commerce personalization at ~200,000 events per second.

- Architected a data lake and warehouse for a global retailer, reducing storage costs; delivered Snowflake marts and automated monitoring.

Solutions Architect, Cloudera

May 2021 – Mar 2022 | Montreal, Canada

- Advised customers adopting Cloudera Data Platform across public, private, and hybrid environments; designed on-premises to AWS migrations and trained engineering teams.

Senior Data Engineer, Intact Insurance

Apr 2020 – May 2021 | Montreal, Canada

- Led data engineers delivering an on-premises to AWS historical-data migration and refactoring pipelines onto Databricks; authored cloud and data-platform best-practice guides.

Earlier: Data Engineer, Ericsson (2019–20) · Data Engineer, CGI (2018–19) · Data Engineer, Synechron (2018) · Big Data Developer, Interglobe (2016–18, Dubai — ~2 PB pipelines) · Software Engineer, MothersonSumi/MIND (2014–16) · Software Developer, NIIT (2012–14).

SELECTED PROJECT

Agentic Expense Manager (open source): Local-first Claude tool-calling agent for web and WhatsApp on a shared service layer with a SQLite source of truth (FastAPI, React); server-side tax calculation, receipt OCR, optional Google Drive/Sheets sync; 345+ backend tests including adversarial sync cases.

CERTIFICATIONS

Snowflake SnowPro Core · AWS Solutions Architect – Associate · AWS Cloud Practitioner · Databricks Certified Associate Developer (Spark 3.0) · Cloudera CCA Spark and Hadoop Developer (CCA-175) · Microsoft Azure Fundamentals (AZ-900)

EDUCATION

Master of Science (M.Sc.), Computer Science — Manav Bharti University (2010–2013)